Baruch COLLEGE

Experience, Conscious Control and Deepfakes in nonverbal communication at FOMC press conferences

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PhD Student's Presentation

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Lord Shiva

What are the top three things people want to hear in an academic paper presentation?

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The top three things people want to hear in an academic paper presentation are:

1. **Clear Research Question & Contribution**: They want to know what problem you're addressing and why it matters. Explain the novelty and significance of your work in the broader academic context.
2. **Key Findings**: Highlight the most important results and insights from your research. Make sure these findings are directly tied to your research question and demonstrate their relevance.
3. **Implications & Impact**: Discuss how your findings contribute to the field, what practical or theoretical implications they have, and any potential for future research or real-world application.

These points allow the audience to quickly grasp the essence, value, and potential impact of your research.

What I intend to do

- Energize the audience
- To be human
- To be of use

How did this project come about?

- Accounting in our minds
- New Things
- Facial Analysis

Background and Context

- Federal Reserve has a branch – Federal Open Market Committee (FOMC) that controls US monetary policy
- Dual Mandate – Price Stability and Maximum Sustainable Employment,
- Financial Stability is an implicit goal
- FOMC meets eight times a year and announces Fed Rate cuts or increases

Background and Context

- Federal Reserve has a branch – Federal Open Market Committee (FOMC) that controls US monetary policy
- Dual Mandate – Price Stability and Maximum Sustainable Employment,
- Financial Stability is an implicit goal
- FOMC meets eight times a year and announces Fed Rate cuts or increases
- Before 2011, these decisions were communicated through official statements
- In 2011, Chair Bernanke introduced FOMC post-meeting press conferences to increase transparency and provide more direct explanations
- Initially, only selected FOMC meetings had press conferences
- This was extended to all FOMC meetings in 2019 under current Chair Powell

For accounting folks

- In accounting research, whatever the CEO does is "researchable"
  - Narcissism is bad (Ham, Seybert and Wang, RAST, 2018)
  - Face traits of CEO (Peng et al., JAR, 2022)
  - Connectedness and Fraud (Khanna, Kim and Lu, JF, 2015)
- In economics research, whatever the Fed Chair/FOMC press conference does is likewise "researchable"
  - What the Fed Chair says – verbal communication (Gomez-cram and Grotteria, JFE, 2022)
  - Informational effects of Fed's policy signals ultimately impact the economy in more ways than just the monetary policy itself (Nakamura and Steinsson, QJE, 2018)
  - Option markets trade before the FOMC announcements (Du, Fung and Loveland, JBF, 2018)

What interaction effects exist in nonverbal communication in FOMC press conferences?

What interaction effects exist in nonverbal communication in FOMC press conferences?

- Nonverbal communication takes up a large portion of our communication (Mehrabian, Book, 1972)
- Investors react negatively to expressions revealed during FOMC press conferences, even when controlling for verbal component (Curti and Kazinnik, JME, 2023)
- Investors react to face-emotion and voice indices during congressional testimonies (Alexopoulos et al., JME, 2023)

What interaction effects exist in nonverbal communication in FOMC press conferences?

My Findings

- Firstly, negative facial expressions affect stock market movements, even when controlling for sentiment of the verbal component
- Secondly, transparent expressions affect markets more than happy, neutral, sad expressions, showing that human expression is a bigger sum than its parts
- Next, fed chairs are affected by recent events, and also show more negative expressions as their term increases. The market also reacts less to the negative expressions, pointing to the fact that market is perhaps getting used to the person
- Lastly, AI is probably not going to replace the Fed Chair due to the individual distinct nature of our faces and expressions

What interaction effects exist in nonverbal communication in FOMC press conferences?

Methodology

- Download FOMC press conferences in the past 10 years
- Split them into 2-second frames
- Use libraries to eliminate frames that do not have Fed Chair's face
- Perform matching of transcript to their video at specific times
- Match the time of the press conference to the market's movement
- Perform a minute by minute average of the expressions
- Collect the macroeconomic variables
- Run regressions

Is this research accounting?

Is this research accounting?

- No?

Is this research accounting?

- Information Transfer
- RAST?

Table 3:

- Do Investors react to negative facial expressions? (Replication of Sophia and Kazzinik, 2023)
- Columns 1,2,3,4,5,6 show that SPY and VIX move in the predicted direction, even when controlling for various aspects of the FOMC press conference

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	%Δ SPY	%Δ SPY	%Δ SPY	%Δ VIX	%Δ VIX	%Δ VIX	%Δ EUR	%Δ EUR	%Δ JPY
Negative Facial	−0.007	−0.008	−0.007**	−0.016	−0.083**	−0.065*	0.003	0.001	0.005**
	(0.005)	(0.006)	(0.002)	(0.043)	(0.040)	(0.033)	(0.003)	(0.003)	(0.002)
Negative Sentiment	0.003*	0.003*	0.003*	0.010	0.009	0.013	0.001	0.001	0.002
	(0.001)	(0.001)	(0.001)	(0.010)	(0.010)	(0.009)	(0.001)	(0.001)	(0.001)
Statement Related	−0.000	−0.000*	−0.000	−0.001	−0.001	−0.000	0.001**	0.001**	0.000
	(0.000)	(0.000)	(0.000)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)	(0.000)
FLS_Ratio	0.004*	0.005*	0.005**	0.005	0.006	0.020	0.001	0.000	−0.000
	(0.003)	(0.003)	(0.002)	(0.017)	(0.017)	(0.015)	(0.001)	(0.001)	(0.001)
Δ FDFD	−0.011	−0.008	0.000	0.088	0.040	0.000	−0.012	0.000	0.007
	(0.018)	(0.018)	(.)	(0.185)	(0.178)	(.)	(0.015)	(.)	(0.013)
MPU	−0.000			0.001	0.000		0.000	0.000	−0.000
	(0.000)			(0.000)	(0.000)		(0.000)	(.)	(0.000)
Predrift SPY	−0.001	−0.001	0.000						
	(0.005)	(0.006)	(.)						
Hawkish	−0.000	−0.000	−0.000	0.002	0.002	0.000	−0.000***	−0.000***	−0.000
	(0.000)	(0.000)	(0.000)	(0.003)	(0.003)	(0.003)	(0.000)	(0.000)	(0.000)
Public_Interest			0.000			0.000	0.001***	0.000	
			(.)			(.)	(0.000)	(.)	
Predrift VIX				−0.006	−0.006	0.000			
				(0.007)	(0.007)	(.)			
Predrift EUR							0.005	0.000	
							(0.005)	(.)	
Predrift JPY									0.010
									(0.011)
Chair FE	No	Yes	No	No	Yes	No	No	No	No
Meeting FE	No	No	Yes	No	No	Yes	No	Yes	No
r^2	0.016	0.290	0.049	0.049	0.065	0.316	0.044	0.235	0.034
N	1389.000	1389.000	1359.000	1359.000	1404.000	1359.000	1404.000	1404.000	1404.000

Table 4:

- Does volume move with negative ? (Replication of Sophia and Kazzinik, 2023)
- Columns 1,2,3,4,6,7 show that there is less trading in the predicted direction, even when controlling for various aspects of the FOMC press conference. This points towards a convergence in traders' beliefs

	(1) SPY Vol	(2) SPY Vol	(3) VIX Tick	(4) VIX Tick	(5) EUR Tick	(6) EUR Tick	(7) EUR Tick
Negative Facial	−0.035	−0.059***	−0.453***	−0.002	0.386***	−0.501***	−0.068
	(0.027)	(0.023)	(0.079)	(0.004)	(0.067)	(0.069)	(0.078)
Negative Sentiment	0.019	0.022	0.004	−0.003	0.168***	0.141***	0.095***
	(0.013)	(0.017)	(0.044)	(0.003)	(0.055)	(0.046)	(0.033)
Statement Related	−0.004*	−0.003	−0.005	0.001	0.028***	0.022***	0.023***
	(0.002)	(0.002)	(0.010)	(0.001)	(0.009)	(0.007)	(0.005)
Hawkish	−0.002	−0.001	0.002	0.000	−0.000	−0.003	−0.007*
	(0.002)	(0.002)	(0.006)	(0.000)	(0.005)	(0.005)	(0.004)
Δ FDFD	0.000	−0.463***	−1.475***	0.000	−1.050***	−1.404***	0.000
	(.)	(0.082)	(0.172)	(.)	(0.213)	(0.180)	(.)
Predrift SPY	0.000	−0.052					
	(.)	(0.041)					
Predrift VIX			−0.105***	0.000			
			(0.009)	(.)			
Predrift EUR					0.697***	0.432***	0.000
					(0.141)	(0.121)	(.)
Chair FE	No	Yes	Yes	No	No	No	Yes
Meeting FE	No	No	No	Yes	No	No	Yes
r^2	0.399	0.122	0.461	0.998	0.069	0.375	0.691
N	1389.000	1389.000	1359.000	1359.000	1404.000	1404.000	1404.000

*, **, *** represent significance at the 10%, 5%, and 1% level.

Table 5:

- Do investors react to transparent reactions?
- Transparent facial expressions show more movement than happy, neutral, sad by themselves, showing that facial interpretation is more complex than its individual components

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	%Δ SPY	%Δ SPY	%Δ SPY	%Δ SPY	%Δ VIX	%Δ VIX	%Δ VIX	%Δ EUR	%Δ EUR
Happy Facial	−0.001*								−0.002***
	(0.001)								(0.001)
Negative Sentiment	0.003**	0.003*	0.003**	0.003**	0.011	0.008	0.013	0.001	0.001
	(0.002)	(0.002)	(0.001)	(0.002)	(0.014)	(0.014)	(0.012)	(0.001)	(0.001)
Hawkish	−0.000	−0.000	−0.000	−0.000	0.002	0.001	0.000	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)	(0.003)	(0.003)	(0.000)	(0.000)
Statement Related	−0.000**	−0.000**	−0.000	−0.000**	−0.001	−0.001	−0.000	0.001**	0.001**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)
%Δ FDFD	−0.012*	−0.011	0.000	−0.017***	0.059	0.026	0.000		
	(0.007)	(0.007)	(.)	(0.007)	(0.066)	(0.065)	(.)		
MPUUS_MPU	−0.000***	−0.000***	0.000	−0.000***	0.000	−0.000	0.000		
	(0.000)	(0.000)	(.)	(0.000)	(0.000)	(0.000)	(.)		
Predrift SPY	−0.002	−0.002	0.000	−0.002					
	(0.002)	(0.002)	(.)	(0.002)					
Neutral Facial		0.003*			0.012				
		(0.002)			(0.013)				
Transparent Facial			−0.006***			−0.068***		−0.002*	
			(0.002)			(0.014)		(0.001)	
Sad Facial				0.007***			0.014		
				(0.002)			(0.023)		
Predrift VIX					−0.006**	−0.006**	0.000		
					(0.003)	(0.003)	(.)		
Predrift EUR								0.003	0.004
								(0.003)	(0.003)
Chair FE	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes
Meeting FE	No	No	Yes	No	No	No	Yes	No	No
r2	0.031	0.032	0.285	0.042	0.042	0.042	0.315	0.315	0.064
N	1389.000	1389.000	1389.000	1389.000	1359.000	1359.000	1359.000	1359.000	1404.000

Table 6:

- Do investors react to expressions that contrast with word sentiment?
- There is no significance for the interacted term, so I take it that the facial expressions are a separate, higher-dimension signaling tool from the content of words.

	(1) %Δ SPY	(2) %Δ SPY	(3) %Δ VIX	(4) %Δ VIX	(5) %Δ EUR
Negative Facial	−0.007	−0.006**	−0.085**	−0.054*	−0.003
	(0.006)	(0.003)	(0.039)	(0.033)	(0.003)
Negative Sentiment	0.004	0.002	0.005	0.014	−0.001
	(0.004)	(0.003)	(0.029)	(0.033)	(0.003)
Negative Facial * Negative_Sent	−0.001	0.000	0.003	−0.002	0.002
	(0.003)	(0.002)	(0.026)	(0.026)	(0.002)
Statement Related	−0.000		−0.001		0.001*
	(0.000)		(0.002)		(0.001)
%Δ FDFD	−0.009		0.035		−0.017
	(0.018)		(0.184)		(0.012)
Predrift SPY	−0.001	0.000			
	(0.006)	(.)			
Predrift VIX			−0.006	0.000	
			(0.007)	(.)	
Predrift EUR					0.002
					(0.005)
Chair FE	No	Yes	Yes	No	Yes
Meeting FE	No	Yes	No	Yes	No
r2	0.022	0.279	0.049	0.315	0.065
N	1389.000	1450.000	1359.000	1420.000	1404.000

Table 7:

- Does the Fed Chair control his/her facial expression?
- As their tenure increases, the negative expressions goes up, while the other expressions go down.
- I take this to mean that they settle more comfortably into their jobs

	(1) Negative	(2) Negative	(3) Negative	(4) Neutral	(5) Neutral	(6) Happy	(7) Sad
Conference Count	0.031***	0.035***	0.000	−0.059***	0.000	−0.023	−0.027
	(0.003)	(0.006)	(.)	(0.013)	(.)	(0.021)	(0.016)
Negative Sentiment	−0.011	−0.016	−0.008	0.030	0.039*	0.037	−0.024
	(0.017)	(0.012)	(0.011)	(0.026)	(0.020)	(0.047)	(0.025)
Statement Related	−0.001	−0.003	−0.000	0.004	0.003	−0.010	0.006
	(0.003)	(0.003)	(0.003)	(0.006)	(0.005)	(0.007)	(0.006)
Δ FDFD	−0.162*	−0.199	0.000	−0.297	0.000	0.175	0.818*
	(0.088)	(0.282)	(.)	(0.340)	(.)	(0.835)	(0.441)
MPU	−0.001***	−0.001	0.000	0.003**	0.000	−0.003**	0.002
	(0.000)	(0.001)	(.)	(0.001)	(.)	(0.001)	(0.001)
Chair FE	No	Yes	No	Yes	No	Yes	Yes
Meeting FE	No	No	Yes	No	Yes	No	No
r2	0.089	0.329	0.558	0.194	0.456	0.064	0.154
N	1404.000	1404.000	1404.000	1404.000	1404.000	1404.000	1404.000

*, **, *** represent significance at the 10%, 5% and 1% level.

Table 8:

- Does experience affect how negative facial expressions are interpreted?
- Firstly, as tenure increases, the expressions become more negative, corroborating Curti and Kazinnik (2023)
- Next, if a congressional testimony has just happened, their negative expressions have larger effect, perhaps showing that Fed Chair get more nervous in the short-term
- (3) and (7) show that Fed chair's career progresses, their negative facial expressions' effects on the market decreases.
- All in all, I suggest that Fed Chairs are more uptight earlier into the careers and then as they settle into the job, they make more negative expressions, and the market interprets them with less volatility

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	%Δ SPY	%Δ SPY	%Δ SPY	%Δ SPY	%Δ SPY	%Δ VIX	%Δ VIX	%Δ VIX	%Δ VIX
Negative Facial	−0.005	−0.000	−0.004	−0.009***	−0.008***	−0.015	0.036	−0.029	−0.079***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.028)	(0.026)	(0.019)	(0.021)
Negative Facial i/r Conference Count	−0.000	−0.001***				−0.005***			
	(0.000)	(0.000)				(0.002)			
Negative Sentiment	0.003*	0.003*	0.003*	0.003*	0.003*	0.009	0.009	0.009	0.009
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.014)	(0.014)	(0.014)	(0.014)
Statement Related	−0.000**	−0.000**	−0.000**	−0.000**	−0.000**	−0.001	−0.000	−0.000	−0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.002)	(0.002)	(0.002)	(0.002)
%Δ FDFD	−0.010	−0.006	−0.010	−0.012*	−0.008	0.066	0.122*	0.148*	0.018
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)	(0.068)	(0.070)	(0.078)	(0.067)
MPUUS_MPU	−0.000		−0.000***	−0.000***	−0.000	0.000	0.001***	0.001***	0.000
	(0.000)		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Predrift SPY	−0.001	0.000	−0.002	−0.002	−0.000				
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)				
Negative * cfquart			−0.002**				−0.023***		
			(0.001)				(0.007)		
Negative * congre 30				0.001				0.038**	
				(0.002)				(0.019)	
Negative * congre 10					0.007*				−0.035
					(0.004)				(0.023)
Predrift VIX						−0.006**	−0.005**	−0.007***	−0.005**
						(0.003)	(0.002)	(0.002)	(0.003)
Chair FE	No	Yes	Yes	Yes	No	Yes	No	No	Yes
r2	0.012	0.028	0.039	0.037	0.014	0.054	0.017	0.013	0.050
N	1389.000	1389.000	1389.000	1389.000	1389.000	1359.000	1359.000	1359.000	1359.000

Panel A: Fed Chair Yellen speaking during a FOMC press conference. Dominant emotion identified by facial analysis software - **Sad**. Complete Analysis: Angry: 0.722%, Disgust: 0.036%, Fear: 21.992%, Happy: 0.057%, Sad: 58.435%, Surprise: 0.021%, Neutral: 18.737%

Panel B: Deepfake of Fed Chair Yellen speaking, using Fed Chair Powell's face, during a FOMC press conference. Dominant emotion identified by facial analysis software - **Angry**. Complete Analysis: Angry: 99.957%, Disgust: 0.0000038%, Fear: 0.001%, Happy: 0.000%, Sad: 0.041%, Surprise: 0.000%, Neutral: 0.000%

Figure 1. Comparison of a deepfake of Fed Chair Janet Yellen during FOMC press conference on September 21, 2016 using Fed Chair Powell and their facial analysis result

Table 9:

- Will the Fed Chair be replaced by AI?
- There is great disparity in the emotions expressed, despite what the program should do, which is simply superimposing the face while preserving the emotions
- Thus, an AI would most likely not replace this human interaction touch.

Emotion	(1) Yellen Original	(2) Yellen Original Overlay Powell	(3) Powell Original	(4) Powell Original Overlay Bernanke	(5) Bernanke Original	(6) Bernanke Original Yellen Overlay
angry	1.903	33.760	78.645	47.689	22.170	4.893
disgust	0.084	5.761	0.022	0.313	0.024	0.354
fear	27.126	1.474	1.316	17.188	5.723	13.929
happy	10.147	3.009	0.141	0.659	2.951	15.799
sad	35.018	43.310	0.169	16.324	27.864	33.051
surprise	1.707	0.225	0.020	0.212	0.272	0.007
neutral	24.014	12.461	19.687	17.615	40.997	31.968

$180 slide

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Ref.: Ms. No. **24-260**
Experience, Conscious Control and Deepfakes in nonverbal communication at FOMC press conferences
Journal of Monetary Economics

6/25/2024

Dear Hunter,

Thank you for submitting your paper, “Experience, Conscious Control, and Deepfakes in Nonverbal Communication at FOMC Press Conferences,” to the Journal of Monetary Economics. We are sorry to report that we will not be able to publish your paper.

Based on our initial review, we find it very unlikely that your manuscript would ultimately make a contribution that would satisfy the referees and our own concerns regarding publication at the Journal of Monetary Economics. The topic of communication by the Fed Chair at post-FOMC press conferences is potentially a good fit for the JME, but we have already published work by Alexopolous et al. on facial expressions of the Fed Chair, so your paper must be judged on its value added relative to that paper. Your paper considers post-FOMC press conferences while theirs considers Congressional testimony, and your paper's methods are slightly different, but by themselves these differences are not sufficient to warrant publication in the JME, in our view. Your results for the short deepfake videos are also unique but raise more questions than answers, in our view--for example, the deepfake videos are supposed to map facial expressions one-to-one between the two speakers, yet your facial expression evaluation code produces very different results on the deepfake video than it does for the original video. This suggests either problems in the deepfake software or problems in your facial expression evaluation code. You might consider just dropping the deepfakes results in future versions of the paper, since they make up only a very small part of the paper, the takeaways from that analysis are not particularly insightful, and those results raise more questions than answers. In the interest of giving you a quick decision, we therefore decided that it is best to stop the editorial process here so that you can turn the paper around quickly and submit it to another journal that is more likely to be a better fit. You might try the JMCB or IJCB.

The JME receives a large number of submissions of which approximately 10% are eventually published. In order to offer authors an expeditious review process, and to focus editorial resources on manuscripts that seem most appropriate in topic and nature of contribution, not all manuscripts are sent out for review to referees. While a fast negative decision with no detailed feedback may be disappointing, it is our hope that authors nevertheless appreciate the opportunity to be able to consider without delay alternative outlets for their work.

Your initial submission fee minus $100 will be refunded to you by Elsevier. Please note that this may take up to 3 months.

Thank you for your interest in Journal of Monetary Economics.

Other Projects I'm working on

Strategic Wash Trading of Bitcoin

Abstract - This paper investigates the phenomenon of wash trading in cryptocurrencies, where traders simultaneously place buy and sell orders to inflate trading volumes artificially. Using high-frequency transaction data from the Mt.Gox Bitcoin exchange, I find that wash trades are strategic in response to low natural trading volume. When wash-trading in an exchange is high, the exchange's activity also mirrors on-chain transactions and other markets' transactions more closely. Wash-trading also takes place more when other asset classes trade less. Additionally, when the information environment is weak and there is more information asymmetry, wash trading's effect is slower but more sticky in the market. Wash traders also take advantage of contemporary events and induce more wash trading to boost volume. The results highlight the finding that wash trades are strategic in nature and this has implications for the financial and legal literature in allowing a better characterization of how wash trades take place, and thus, how they can be detected.

Why is it so hard to find a job now? Enter Ghost Jobs

Abstract - This study investigates the emerging phenomenon of "ghost hiring" or "ghost jobs", where employers advertise job openings without intending to fill them. This causes significant disruptions in the labor market. Using a novel dataset from Glassdoor and employing advanced natural language processing techniques, I find that up to 21% of job ads may be ghost jobs, and this is particularly prevalent in specialized industries and in larger firms. The trend could be due to the low marginal cost of posting additional job ads and to maintain a pipeline of talents. The results show that policy-makers should be aware of such a practice as it causes significant job fatigue and distorts market signals. I also caution against empirical studies which use job-ads directly as a significant portion may be ghost jobs.

BigBird for ESG Reports - A Sparse Attention Model For Differences Between Languages In Textual Analysis

Abstract - This paper introduces BigBird-ESG, a domain-specific transformer architecture pre-trained on manually classified paragraphs from Chinese ESG reports between 2016 to 2020. Our results show that BigBird-ESG, with its sparse attention mechanisms, more efficiently processes Chinese ESG reports due to innate qualities of the Chinese language and its difference from the English language. We show that BigBird-ESG outperforms BERT and FinBERT under specific conditions in both sentiment and category classification tasks. The findings suggest that language specificity affects the accuracy of LLM models in parsing textual data. Our findings affect the future advancement of multi-modal LLMs and transfer learning, which should consider language-specific qualities when interpreting sources of financial information. We also use state-of-the-art OCR to coherently extract paragraphs from the ESG reports, which preserves the meaning of the textual data and we use this new methodology to show that tone in Chinese ESG reports is correlated with ESG ratings.

Table Extraction using Data Augmentation and Path Aggregation Network on ESG Reports using pure OCR

Abstract - Tables are widely used to present information and there has been many existing commercial and open-source techniques. Such tabular data can exist in (1) non-raster format, where the data does not have a grid of individual pixels but are instead composed of paths with machine-readable text, or (2) raster format, which are pixel-based images pasted in the document. Non-raster tables can be picked up by PDF text-parsing tools such as Tabula and PDFplumber because they exist as machine-readable texts but even then, the accuracy depends heavily on the formatting of the table and the deep learning model used in the parsing tool. As such, research that involves extracting information ESG reports often just parses the entire ESG report for textual data and returns the tabular information embedded in the text, which messes up the information (Bast and Korzen, 2017).This paper proposes a data augmentation method with Path Aggregation Network (PANet)

Thank you to the PhD Committee for taking a chance on me